

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First Semester of M. Tech (EC)(EVD) Examination****March-April 2018****EC749 CMOS VLSI DESIGN-I****Date: 31.03.2018, Saturday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below.**

- a. Define the following terms for logic gate. **[04]**
Critical Path, Positive Slack, Setup Time, Hold Time.

- b. Draw and explain the current-voltage characteristics of an n-channel MOS transistor **[03]**
including the channel length modulation effect.

Q - 2.a What is the lower limit of supply voltage of a CMOS inverter? What happens if the **[04]**
supply voltage is further reduced?

Q – 2.b Answer any two questions. **[10]**

- (i) Prove that the transmission gate gives nearly constant equivalent resistance as a function of output voltage
- (ii) Implement the following function using CMOS logic gates.
 $Z = [A(D+E)+BC)]'$
Draw the optimized stick diagram for the same.
- (iii) Compare the two technology scaling methods, namely, (1) the constant electric-field scaling and (2) the constant power-supply voltage scaling. In particular, show analytically by using equations how the delay time, power dissipation and power density are affected in terms of the scaling factor, S .

Q – 3.a Explain the CMOS transistor fabrication flow in detail. **[04]**

Q-3. b Answer any two questions. **[10]**

- (i) By considering two cases $V_{in} = V_{OL}$ (Low) and $V_{in} = V_{OH}$ (High), explain power consumption and chip area in case of resistive load inverter.
- (ii) What are the advantages of CMOS dynamic logic? Compare the NORA CMOS logic and Zipper CMOS logic.

- (iii) “The average power dissipation of the CMOS inverter is proportional to the switching frequency f .” Justify the above statement with proper diagram and explanation.

SECTION – II

Q - 4 Answer the questions below.

- a. Draw the energy band diagram of the MOS transistor. [02]
 - b. Implement the following boolean function: [03]
 - (i) $F_1 = [(A+B)(C+D)]'$ using minimum CMOS logic gates.
 - (ii) $F_2 =$ Two input Multiplexers using transmission gates.
 - (iii) $F_3 = (AB)'$ using CPL family.
 - c. Indicate True or False and justify your answer with suitable diagram. CMOS NAND2 [02]
gate circuit consists of a series-connected n-net and a parallel-connected complementary p-net.
- Q – 5.a** Find the worst-case rising and falling propagation delays of a 2-input NAND gate with [05]
transistor widths chosen to achieve effective rise and fall resistances equal to a unit inverter.

OR

- Q – 5.a** What is the threshold voltage of a CMOS inverter? Derive the expression $(W/L)_p = 2.5$ [05]
 $(W/L)_n$ for symmetric CMOS inverter.
- Q – 5.b** What is difference between latch and flip-flop? Explain the JK latch circuit and its [05]
CMOS realization.
- Q – 5.c** Explain the operation of DRAM cell with suitable diagrams. [04]

OR

- Q – 5.c** Draw the RC delay model for a CMOS inverter. Prove that layout style has an impact [04]
on total delay calculation.
- Q - 6.a** Draw and Explain two-input depletion – load NOR gate and derive the V_{OL} equation [06]
for the same.
- Q – 6.b** Explain the following blocks with necessary diagrams: [06]
- (i) Shift Registers
 - (ii) Queues
- Q – 6.c** What is the need of dynamic voltage bootstrapping? [02]

OR

- Q – 6.a** Explain conditions of accumulation, depletion and inversion in MOS structure using neat and clean diagram. **[06]**
- Q – 6.b** Design a complementary logic circuit for $F(ABCD) = \sum m(0,1,2,3,4,5,6,7,8,9)$. **[05]**
Implement the same using precharge and evaluate logic.
- Q – 6.c** What is Logical Effort? Find the Logical effort for Inverter. **[03]**